Sungsoo Kim

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Research Interests

My current interests include **Scattering Amplitudes and Particle Physics**.

Education

Seoul National University, BS in Physics *(Valedictorian)* Mar 2019 - Feb 2026

- Summa Cum Laude
- Military Service: Aug 2021 - Feb 2023
- **Selected Graduate Courseworks:** Quantum Field Theory, General Relativity, Classical Mechanics

Daejeon Science High School For The Gifted, High School Diploma Mar 2016 - Feb 2019

- Sub Member of the Korean Team at **International Physics Olympiad 2018**

Publications

[1] Classical eikonal in relativistic scattering Nov 2025

S. S. Kim*, H. J. Lee* S. M. Lee*

JHEP **11** (2025) 032

[2] Classical Eikonal from Magnus Expansion Jan 2025

J. H. Kim*, **S. S. Kim***, J. W. Kim* S. M. Lee*

JHEP **01** (2025) 111

*Equal contribution (co-first authors)

Experience

Completed Projects

Classical Eikonal in Relativistic Scattering Ended Aug 2025

Collaborators: Sungsoo Kim*, Hojin Lee*, Sangmin Lee*

- Established an explicit diagrammatic correspondence between the Magnus expansion and WQFT, clarifying how non-rooted trees and causal weights arise in relativistic kinematics.
- Analyzing rigorously the standard WQFT setup with two straight–line worldlines interacting by electromagnetic or gravitational fields.
- Demonstrated that the eikonal framework uniformly treats particles and fields, enabling radiation observables—such as Compton and 1.5PM emission—to be derived from the same generating functional.

Classical Eikonal from Magnus Expansion Ended Nov 2024

Collaborators: Joon-Hwi Kim*, Jung-Wook Kim*, Sungsoo Kim*, Sangmin Lee*

- Reformulated the classical eikonal phase as a Magnus expansion of the time-evolution operator connecting in- and out-states.
- Proved this expansion holds for all tree-level diagrams and identified the coefficient structure associated with directional propagators.
- Contributed to extending the Murua formula to general tree-level diagrams and clarifying its role in computing diagrammatic weights.
- Implemented the complete expansion algorithm in `Mathematica` and, **at the Wolfram team's suggestion, uploaded the example file to the Wolfram Community as an exemplar of research-grade `Mathematica` usage.**

*Equal contribution (co-first authors)

CERN Summer Student Programme 2023 Jun 2023 – Aug 2023

Summer Student, CERN

Project: *Experimental Analysis of Crucial Decays with Potential to Violate the Standard Model*

Supervisors: Chen Chen, Jinlin Fu

- **Project Paper:** Feasibility Investigation into Detecting $D_s^+ \to \tau^+ + \nu_\tau$ in pNe Collisions at LHCb [Link to Paper].
- **Presentation:** Presented findings at the LHCb Summer Student Presentation on August 29, 2023.
- Selected as **one of two representatives** from Korea for the fully funded CERN Summer Student Programme.
- Gained hands-on experience in experimental research within the LHCb experiment, including understanding current experimental processes and challenges.

Scholarships & Awards

Presidential Science Scholarship Mar 2021 – Present

- Provided full tuition and additional benefits regardless of the university.
- Selected as one of 20 prospective students nationwide in Physics.

Merit-based Scholarship (Full Tuition Waiver, 3 Semesters) Fall 2019 – Fall 2020

- Granted to outstanding students, covering full tuition fees; awarded only to top achievers.

Dean's List (4 Semesters)

- Awarded to top-performing students for exceptional academic excellence.

Extracurricular Experience

GLEAP Mar 2021 – Present

Accredited Group of Outstanding Undergraduate Students
College of Natural Sciences, Seoul National University

- Collaborated with the education offices of Seoul and Busan to give natural science lectures to high school students.
- Engaged in intellectual exchanges not only within several members in this group but also with other institutions.

Language & Skills

Languages: Korean (Native), English (Intermediate)

Skills: `Mathematica`, LaTeX, Python, MATLAB